

## Case 04: Design and Development of a Patient-Specific Acetabular Cage Using SolidWorks

### 1. Introduction

An acetabular cage is an orthopedic implant used to reconstruct large acetabular defects caused by trauma, tumor resection, revision total hip arthroplasty, or severe bone loss. The cage provides mechanical stability, supports the acetabular component, and facilitates fixation to the surrounding pelvic bone using screws.

In complex cases, patient-specific acetabular cages are designed from CT data to match the patient's anatomy and improve implant fixation.

### 2. Need for Patient-Specific Acetabular Cage

- Management of severe acetabular bone defects
- Pelvic reconstruction after tumor resection
- Revision hip replacement surgery
- Restoration of hip biomechanics
- Improved implant fixation
- Better anatomical fit

### 3. Objective

- To design a patient-specific acetabular cage using SolidWorks.
- To understand the structural features of acetabular reconstruction implants.
- To create a CAD model suitable for additive manufacturing.
- To study fixation strategies used in pelvic reconstruction.

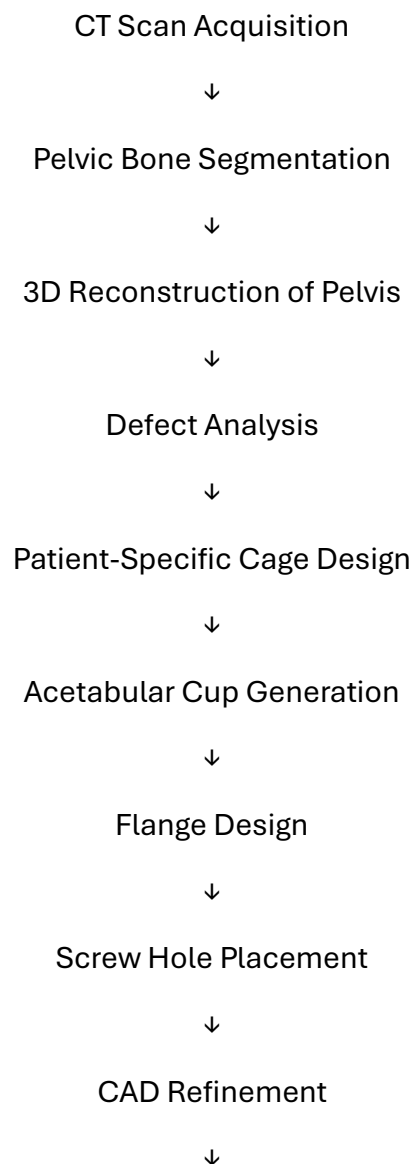
### 4. Software Used

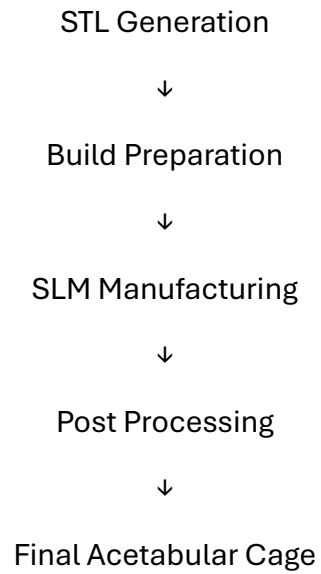
| Component                      | Software                      |
|--------------------------------|-------------------------------|
| CT Segmentation (PSI Workflow) | D2P / 3D Slicer               |
| Anatomical Modeling            | Geomagic Freeform / Meshmixer |
| Freeform Implant Design        | Meshmixer / Freeform          |

|                               |                    |
|-------------------------------|--------------------|
| <b>CAD Refinement</b>         | SolidWorks         |
| <b>Build Preparation</b>      | Materialise Magics |
| <b>STL Generation</b>         | SolidWorks         |
| <b>Additive Manufacturing</b> | SLM                |

For Patient-Specific Implant (PSI) development, defect reconstruction and freeform anatomical modeling are typically performed using **Geomagic Freeform** or **Autodesk Meshmixer**, while final engineering refinement and screw-hole design can be completed in SolidWorks.

## 5. Methodology





## 6. Design Features

| Feature              | Function                        |
|----------------------|---------------------------------|
| Acetabular Cup       | Supports acetabular liner       |
| Iliac Flange         | Fixation to ilium               |
| Ischial Flange       | Additional pelvic fixation      |
| Multiple Screw Holes | Enhanced primary stability      |
| Anatomical Contour   | Improved fit to patient anatomy |
| Open Structure       | Reduced implant weight          |

## 7. Materials Used

| Component        | Material       | Reason for Selection                         |
|------------------|----------------|----------------------------------------------|
| Acetabular Cage  | Ti-6Al-4V ELI  | High strength and excellent biocompatibility |
| Fixation Screws  | Titanium Alloy | Corrosion resistance and stability           |
| Acetabular Liner | UHMWPE         | Low friction articulation                    |
| Femoral Head     | CoCrMo Alloy   | Excellent wear resistance                    |

## 8. Design Specifications of Patient-Specific Acetabular Cage

| Parameter                | Specification                      |
|--------------------------|------------------------------------|
| Implant Type             | Patient-Specific Acetabular Cage   |
| Design Method            | CT-Based Anatomical Reconstruction |
| Material                 | Ti-6Al-4V ELI                      |
| Manufacturing Method     | Selective Laser Melting (SLM)      |
| Fixation Method          | Multiple Locking Screws            |
| Screw Hole Configuration | Multi-Directional                  |
| Anatomical Features      | Iliac and Ischial Flanges          |
| Implant Geometry         | Patient-Specific                   |
| File Format              | STL                                |

## 9. Importance of the Implant

| Clinical Importance                  | Benefit                            |
|--------------------------------------|------------------------------------|
| Pelvic Reconstruction                | Restores structural integrity      |
| Hip Joint Stability                  | Prevents implant loosening         |
| Load Transfer                        | Improves biomechanical performance |
| Patient-Specific Fit                 | Reduces surgical adjustments       |
| Screw Fixation                       | Provides immediate stability       |
| Additive Manufacturing Compatibility | Enables complex geometries         |

